

ABHISEK MISHRA

Senior Solution Architect | Engineering Leader | AI & Fintech Domain Expert

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OVERVIEW

Visionary **Solution Architect and Engineering Leader** with over **13 years** of experience building mission-critical distributed systems for the banking, government, and fintech sectors. Currently pioneering the development of autonomous, **Agentic AI workflows using LangGraph and Temporal** to ensure durable execution in production environments. Expert in architecting high-throughput microservices and event-driven platforms that bridge the gap between traditional enterprise stability and generative AI innovation. Recognized as an **"IT Legend"** for technical excellence and driving large-scale digital transformation through high-performing engineering squads.

PRIMARY SKILLS

Core AI & Architectural Competencies

- **AI Orchestration:** Expert in **Multi-Agent Systems, LangGraph**, and LLM integration
- **Advanced AI Engineering:** Specialist in **Model Context Protocol (MCP)** for connecting LLMs to secure banking & compliance data sources.
- **Durable Execution Resiliency:** Mastery of **Temporal.io** and production-grade error recovery systems (Circuit Breakers, Exponential Backoff, Checkpoint-based Recovery).
- **Architecture:** Event-Driven Microservices, Domain-Driven Design (DDD), Service Resiliency, and Scalable Solution Design.
- **AI-Assisted Engineering:** Advanced proficiency in **Claude Code and Cursor** to accelerate architectural prototyping, automated refactoring, and developer velocity.
- **QA Automation:** Implementing **Playwright MCP** for autonomous UI automation and **AI-driven code reviews** via custom Git hooks.
- **Tech Stack:** Node.js, TypeScript, MongoDB, Redis, GraphQL, Azure, OpenShift.

Strategic Leadership & Influence

- **Architectural Advocacy:** Proven ability to translate complex distributed systems and AI agent workflows into actionable roadmaps for business stakeholders.
- **Technical Multiplier:** Acted as a senior coach for cross-functional engineering teams, standardizing best practices for API design, logging frameworks, and service resiliency.
- **Modern SDLC Innovation:** Engineered **Claude Code plugins** to automate the migration of legacy services to modern stacks, reducing refactoring time by 40%.
- **Rapid Innovation:** Utilizing **Cursor and Claude Code** to rapidly build and demonstrate architectural Proof-of-Concepts (PoCs), drastically reducing the feedback loop with stakeholders.

FEATURED AI PORTFOLIO PROJECTS

Unified RFI Management System

Emirates NBD

- **Architectural Role:** Designed a production-grade, multi-agent platform to automate high-stakes banking **Requests for Information (RFIs)** for Sanctions Screening, AML, and KYC.
- **Innovation:** Orchestrated **6 specialized agents via LangGraph** and implemented **3 MCP Servers** to provide 15+ secure banking and compliance tools to the LLM.
- **Enterprise Resiliency:** Engineered a "Safety-First" architecture featuring **Circuit Breakers** for 8 external services and automated checkpoint recovery for long-running workflows.
- **Autonomous Testing:** Integrated Playwright MCP to perform automatic UI verification and regression testing of the RFI dashboard.

Autonomous Video Synthesis Engine

- **Solution Design:** Architected a distributed engine for automated video clip generation using **Agentic AI and Durable Execution** frameworks.
- **Reliability:** Utilized **Temporal** to manage state and recovery for complex, asynchronous media processing tasks.
- **AI Governance:** Implemented **AI-powered code review Git hooks** to ensure all agent-logic updates adhere to strict architectural and security standards.

EXPERIENCE

Lead Software Engineer

Emirates NBD

July 2019 – Ongoing Dubai, UAE

Full Stack Engineer

Xebia Inc.

Dec 2018 – June 2019 Abu Dhabi, UAE

Full Stack Engineer

Tata consultancy services

March 2014 – Dec 2018 Bhubaneswar, Odisha, India

Frontend Engineer

Mindfire Solutions

July 2013 – March 2014 Bhubaneswar, Odisha, India

PROJECTS

Tablet Banking, ATM, CRM, and more

Emirates NBD

- **Architected a highly scalable, event-driven microservices platform** to underpin the bank's omnichannel initiatives. The system leveraged Node.js for high-throughput APIs, MongoDB for flexible data modeling, and Redis for distributed caching, specifically designed to support the seamless integration of Tablet Banking, ATM servicing, and CRM systems.
- **Championed the adoption of advanced architectural principles**, including Domain-Driven Design and Service Resiliency, to decouple core business capabilities. This strategic shift not only enhanced platform stability and fault tolerance but also significantly improved developer velocity and time-to-market for new financial products.
- **Led the entire solution lifecycle** for multiple modules, from requirement analysis and technical design to deployment, monitoring, and post-launch support. This demonstrated command of end-to-end solution delivery and operational excellence.
- **Served as a go-to backend lead for cross-squad collaborations**, acting as a technical facilitator between architects, product owners, and QA to align technical execution with strategic objectives.
- **Exercised technical leadership and influence to elevate the overall quality and efficiency** of multiple engineering teams by mentoring backend engineers and defining engineering best practices across squads, including logging, API standards, and error handling frameworks.

TAMM - Honour the dead

Abu Dhabi government

- **Architected and developed microservices** to interact with various government entities using TypeScript, ensuring secure and scalable communication between systems.
- Developed reusable front-end components using ReactJS to enforce design consistency and accelerate development cycles.

Precious metals online

CIBC

- **Led the end-to-end solution design and development of a new digital financial product** for purchasing precious metals. The architecture focused on secure, scalable API design to manage complex financial transactions and seamlessly integrate with third-party payment gateways and authentication services.
- Designed the database architecture for the solution using MongoDB and Mongoose.

Richie brothers auctioneers

CIBC

- Developed an interactive User Interface using **Angular 4** and **Bootstrap**.
- Developed RESTful Web services in NodeJS and expressJS to integrate user facing angular components with server side.
- Applied Angular's template driven validations for client side validations and CSS for further enhancements of the presentation layer.
- User authentication through JSON Web Token (JWT).
- Integrated **mongoDB** database using **mongoose**.

Foreign exchange centre

CIBC

- **Contributed to the front-end architecture and user experience design for an international remittance application.** Focused on creating a responsive and intuitive interface that simplified complex financial transactions, while ensuring a secure and compliant experience.
- Worked on UI pages using AngularJS, Bootstrap, AJAX and JSON.

EDUCATION

B-Tech in Electronics and Telecommunication

Biju Patnaik university of Technology

📅 Aug 2009 – May 2013

AWARDS AND ACHIEVEMENTS

- Received IT Legend award from **Emirates NBD** 2 years in a row for **Q4 2021** and **Q3 2022**.
- Received **On the Spot** award on various occasions.
- Certified by **Microsoft** as **HTML5, CSS3, Javascript developer**.